

ASSESSMENT TOOL

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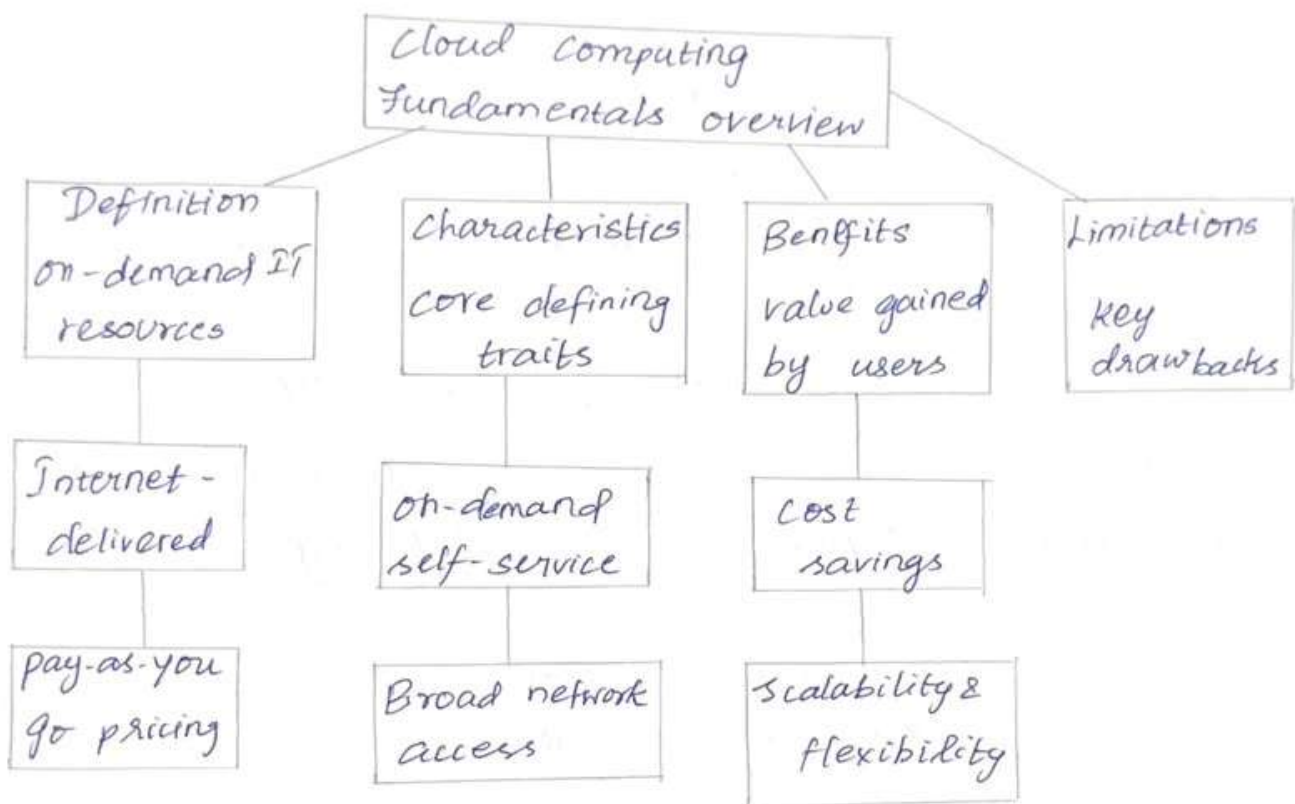
RegNo: 192524059

Course Name: cloud computing and big
Data Analytics

Course Code: CSA1507

Concept Mapping

Concept Map 1: Cloud Computing Fundamentals. Included definitions, characteristics, benefits, and limitations.



This concept map shows cloud computing branching into four pillars, each with two representative sub-points:

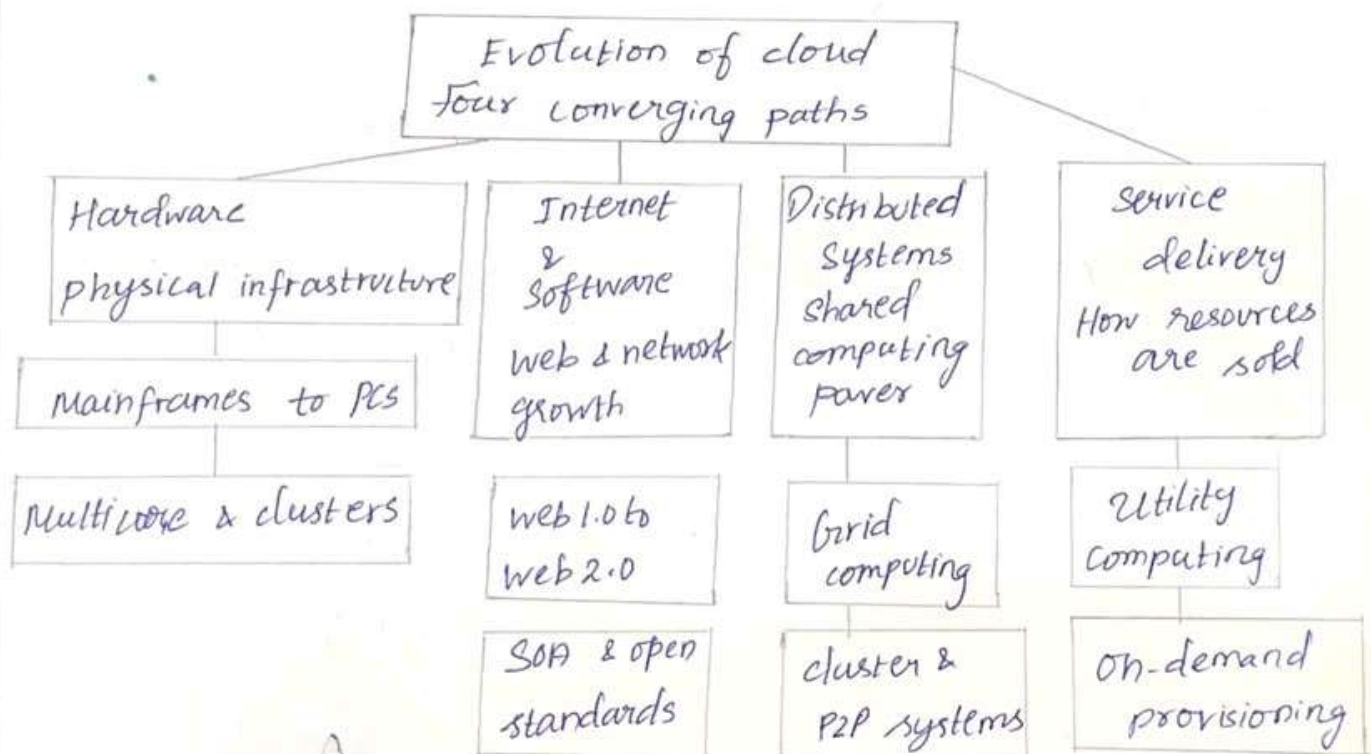
Definition - cloud computing as internet delivered IT services billed on as pay-as-you go basis.

Characteristics — the NIST-style traits like on demand self-services and broad network access (other include resource pooling, rapid elasticity, and measured service)

Benefits — cost savings and scalability/flexibility (also worth noting: disaster recovery, automatic updates, and global accessibility).

Limitations — security/privacy risks and vendor lock-in (also relevant: downtime dependency on internet connectivity and compliance complexity).

Concept Map 2: Evolution of cloud computing. Connect hardware evolution, internet software evolution, distributed systems, service delivery.



Hardware - physical infrastructure

- Mainframes to PCs
- Multicore & clusters

Internet & software - web & network growth

- web 1.0 to web 2.0
- SOA & open standards

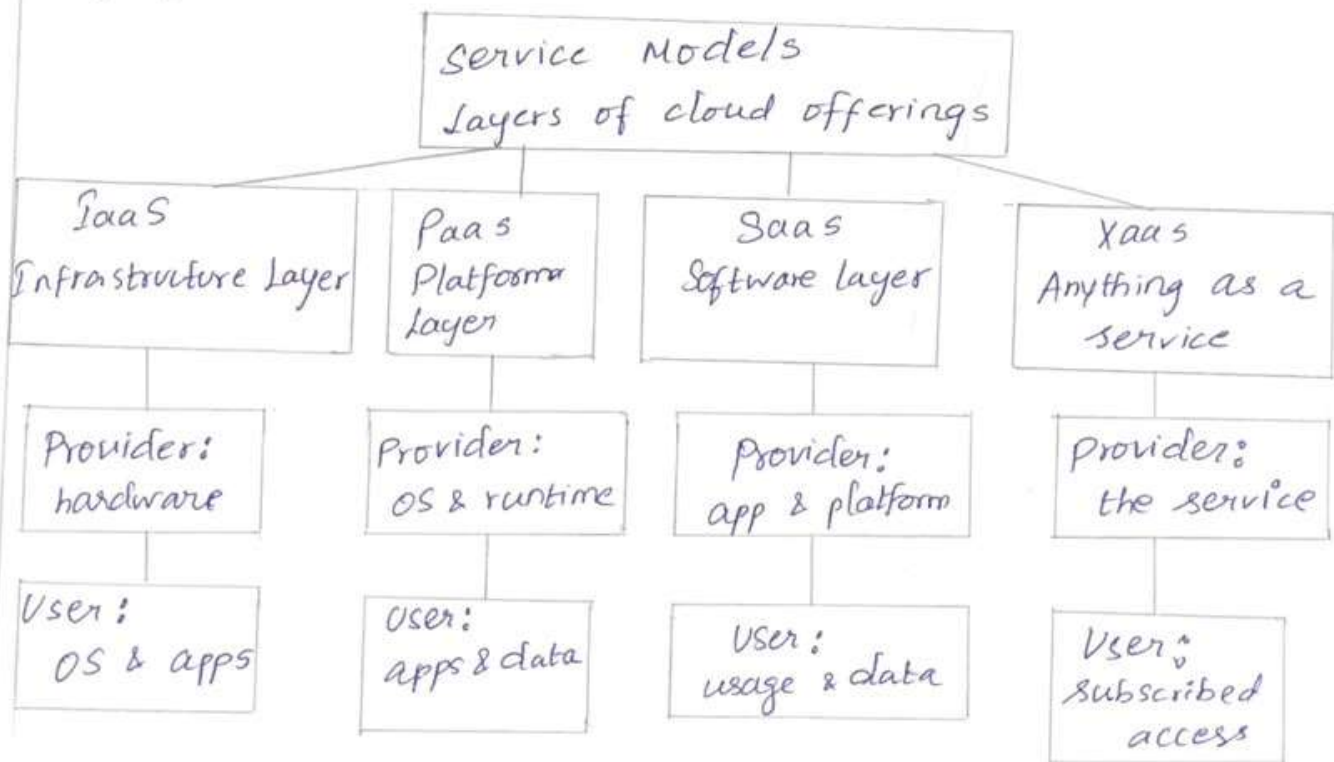
Distributed systems - shared computing power

- Grid computing
- Cluster & P2P systems

Service delivery - how resources are sold

- Utility computing
- On-demand provisioning

Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, XaaS with provider responsibilities and user responsibilities.



cloud service model

✓ IaaS

- Provider responsibility → physical hardware

- User responsibility → OS & applications

✓ PaaS

- Provider responsibility → OS & runtime

- User responsibility → apps & data

✓ SaaS

- provider responsibility → app & platform

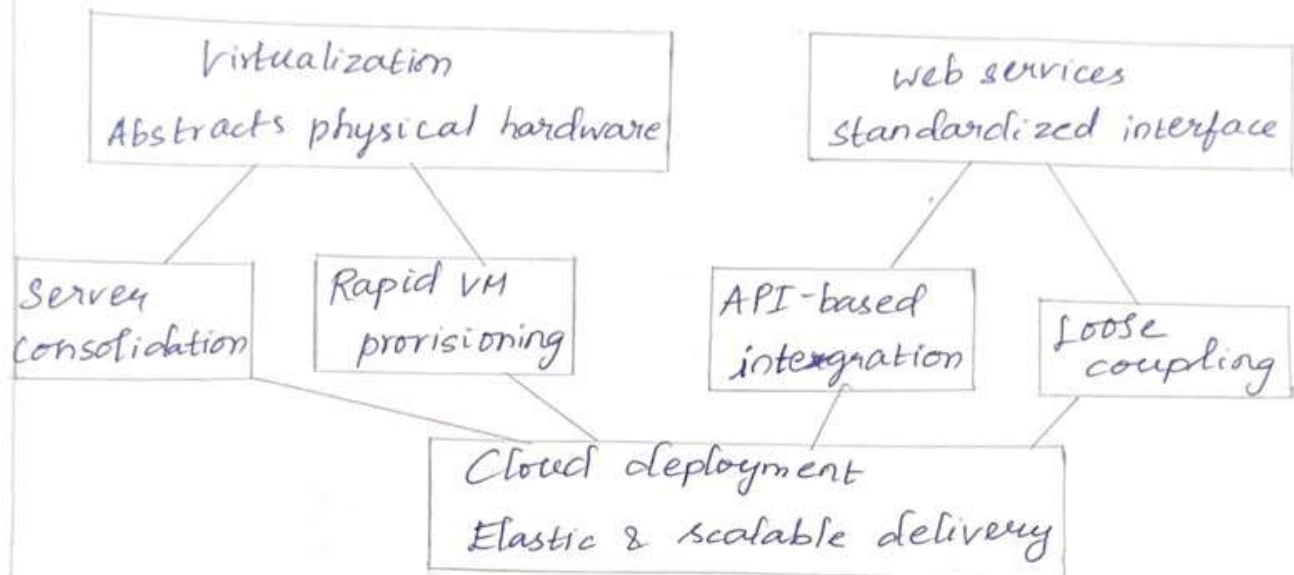
- User responsibility → usage & data

✓ XaaS

- provider responsibility → specific service

- User responsibility → subscribed access

Concept Map 4: Virtualization and web services as Enablers of cloud. show how these two concepts support cloud deployment and scalability.



Enablers of cloud deployment & scalability

virtualization - abstracts physical hardware

- ✓ Server consolidation, Rapid VM provisioning

web services -

Standardized interfaces

- ✓ API-based integration
- ✓ Loose coupling

converges into → cloud deployment (elastic & scalable delivery).